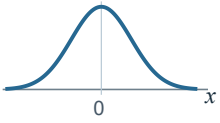
(a) Fixed incident beam

$E_{in}(x, y)$: spatial amplitude and phase



8.2 keV; FWHM = 7.11 μm
Sample 0.25 m downstream of the beam waist

(b) Smooth and translate the wire

Cu cylinder: a = 5 μm; axis parallel to y

$t_\sigma = G_\sigma * t_0$, $\sigma_L = \sigma_R = 4$ nm

$t_n(x, y) = t_\sigma[x - x_0^{(n)}]$

43 offsets: −10.0 to +10.16 μm; step 0.48 μm

(c) Construct the sample-exit field

Interpolate attenuation and phase onto the beam grid

$$E_{exit}^{(n)}(x, y) = E_{in}(x, y)\exp[-\mu t_n(x, y)/2]\exp[-ik\delta t_n(x, y)]$$

(d) Propagate to the detector with the Fresnel approximation

Sample-plane quadratic phase

$$U_n = E_{exit}^{(n)}\exp[ik(x^2 + y^2)/(2Z)]$$

→ **2D FFT** →

$$\tilde{U}_n(q_x, q_y)$$

Detector intensity

$$I_D^{(n)} \propto |\tilde{U}_n(kX/Z, kY/Z)|^2$$

Z = 1.31 m; 4096 × 4096 grid; $\Delta x = \Delta y$ = 8 nm; $\Delta X = \Delta Y \approx$ 6.04 μm

(e) Extract the SAXS profile and quantify asymmetry

Average 10 central rows: $\bar{I}_n(q_x)$

Use $q_x = kX/Z$; exclude $|q_x - q_c| \le q_{min}$

$q_{min} = 0.10\ \text{nm}^{-1}$; $q_c = 0$ or $+0.005\ \text{nm}^{-1}$

LEFT	EXCLUDED	RIGHT

→

Integrate left and right regions

$$S_L = \Sigma_{left}\,\bar{I}_n,\ S_R = \Sigma_{right}\,\bar{I}_n$$

$$f_L^{(n)}(q_c) = \frac{S_L^{(n)}(q_c)}{S_L^{(n)}(q_c) + S_R^{(n)}(q_c)}$$

Repeat for every wire offset